BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT30** **Date: 10-01-2026**

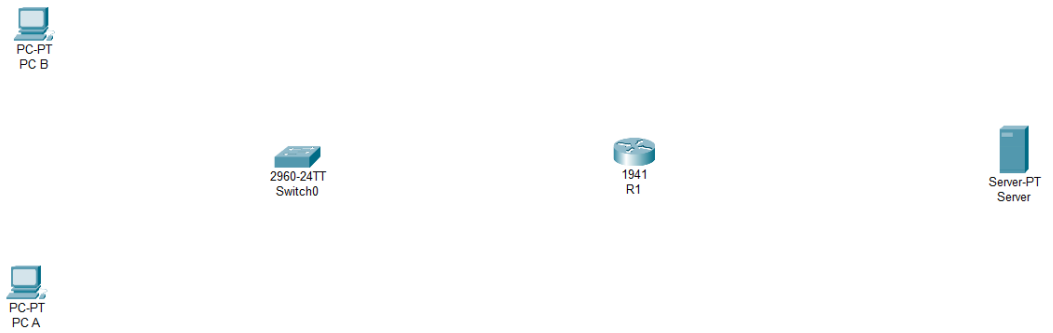
Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number:- 07

Q. Configure a Zone-Based Policy Firewall.

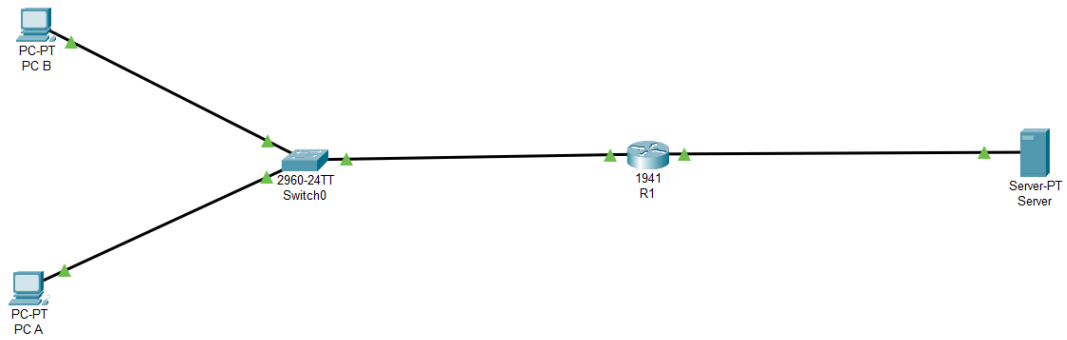
Steps:

1. Open the Cisco packet
2. Add the two PCs(PCA, PCB), one switch of 2960-24TT, one router of 1941(R1), and one server



3. Click on R1 router → Go to the "Physical" tab → Turn off the router → Drag and drop the "HWIC-2T" module into an empty slot → Turn the router back on.

4. Now connect copper straight-through wire to devices as given below.
   - **PCA → Switch:** PCA – FastEthernet0 → Switch FastEthernet0/1
   - **PCB → Switch:** PCB – FastEthernet0 → Switch FastEthernet0/2
   - **Switch → Router:** Switch FastEthernet0/3 → R1 GigabitEthernet0/0
   - **Router → Server:** R1 GigabitEthernet0/1 → Server FastEthernet0



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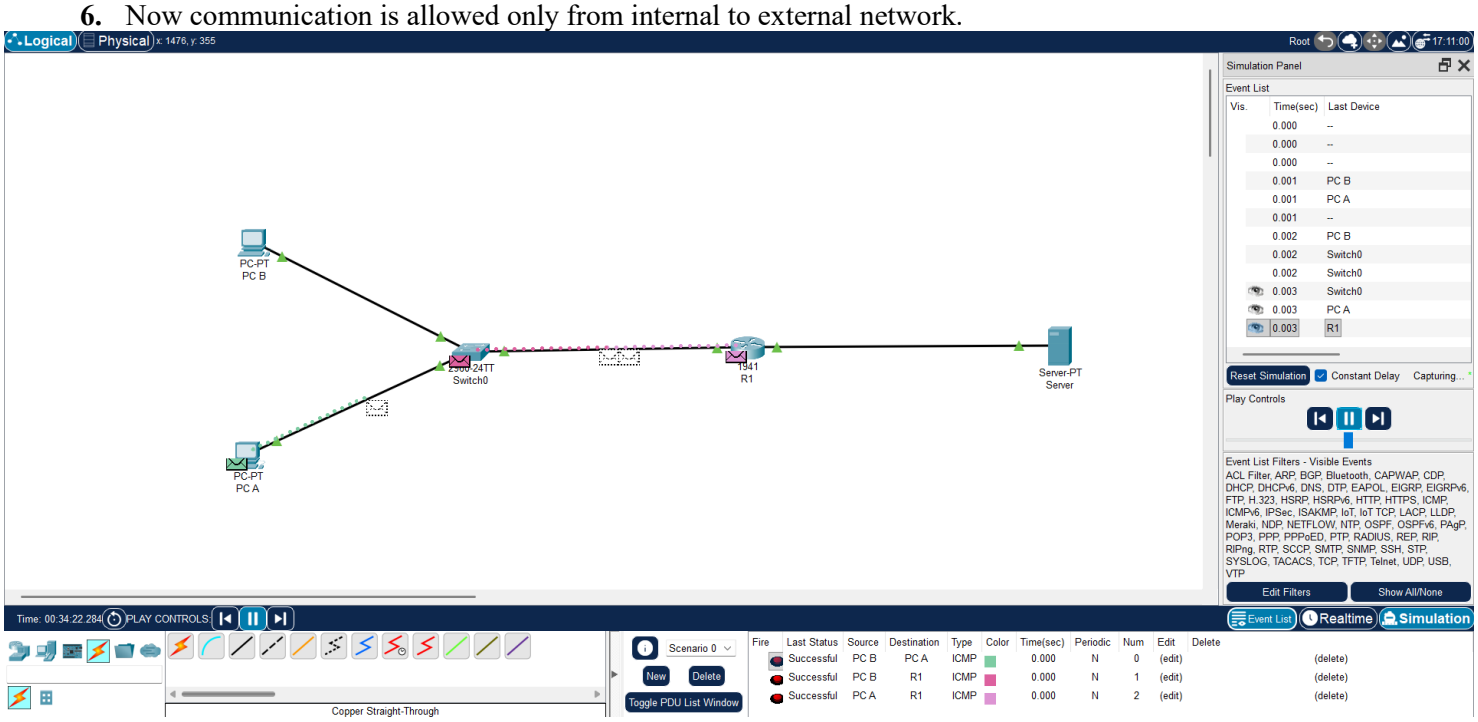
5. Assign the addresses as follow:
   - To assign addresses to Server, PCA and PCB go to the desktop → IP configuration.
   - For R1 go to config.

IP Config	IPv4 Address	Subnet Mask	Default Gateway
Server	10.10.10.2	255.255.255.252	10.10.10.1
PCA	192.168.0.3	255.255.255.0	192.168.0.1
PCB	192.168.0.2	255.255.255.0	192.168.0.1
R1 G0/0	192.168.0.1	255.255.255.0	-
R1 G0/1	10.10.10.1	255.255.255.252	-

6. Now communication is allowed only from internal to external network.



7. Click on R1 → CLI → paste these lines in that, then close.
   - enable secret enpa55
   - line console 0
   - password conpa55
   - login
   - exit

   - ip domain-name ccnasecurity.com
   - username admin secret adminpa55
   - line vty 0 4
   - login local
   - exit

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- crypto key generate rsa
- yes
- 512

```
R1(config)#crypto key generate rsa
% You already have RSA keys defined named R1.ccnasecurity.com .
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#
```

- ex
- show version

R1

Physical Config CLI Attributes

```
cisco1941 uptime is 59 minutes, 17 seconds
System returned to ROM by power-on
System image file is "flash0:c1900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:

License UDI:

-------------------------------------------------
Device#   PID                   SN
-------------------------------------------------
*0        CISCO1941/K9          FTX15243Q1P-


Technology Package License Information for Module:'c1900'

----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102


R1#
R1#
```

Teacher's Signature

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- license boot module c1900 technology-package securityk9
- y

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1(config)#license boot module c1900 technology-package securityk9
*Mar 1 1:3:8.266: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:3:8.266: %SSH-5-ENABLED: SSH 1.5 has been enabled
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
together with an additional  payment.  You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the  product,  including  during the 60 day  evaluation  period,  is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day  evaluation  period,  your  use of the  product  feature will be
governed  solely by the Cisco  end user license agreement (link above),
together  with any supplements  relating to such product  feature.  The
above  applies  even if the evaluation  license  is  not  automatically
terminated  and you do  not receive any notice of the expiration of the
evaluation  period.  It is your  responsibility  to  determine when the
evaluation  period is complete and you are required to make  payment to
Cisco for your use of the product feature beyond the evaluation period.

Your  acceptance  of  this agreement  for the software  features on one
product  shall be deemed  your  acceptance  with  respect  to all  such
software  on all Cisco  products  you purchase  which includes the same
software.  (The foregoing  notwithstanding, you must purchase a license
for each software  feature you use past the 60 days evaluation  period,
so  that  if you enable a software  feature on  1000  devices, you must
purchase 1000 licenses for use past  the 60 day evaluation period.)

Activation  of the  software command line interface will be evidence of
your acceptance of this agreement.


ACCEPT? [yes/no]: y
% use 'write' command to make license boot config take effect on next boot

R1(config)#
```

- zone security internal
- ex
- zone security external
- ex
- ex
- show version

Teacher's Signature

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R1

Physical Config CLI Attributes

```
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Last reload type: Normal Reload

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A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
 --More--
%SYS-5-CONFIG_I: Configured from console by console

If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
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License Info:

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Device#   PID                   SN
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Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102


R1#
```

- conf t
- ip access-list extended 101
- permit ip 192.168.0.0 0.0.0.255 any
- ex

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip access-list extended 101
R1(config-ext-nacl)#permit ip 192.168.0.0 0.0.0.255 any
R1(config-ext-nacl)#ex
R1(config)#
```

- conf t
- ip access-list extended 101
- permit ip 192.168.0.0 0.0.0.255 any
- ex
- class-map type inspect match-all 101
- match access-group name 101
- exit
- policy-map type inspect 101
- class type inspect 101
- inspect

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- exit
- exit
- zone-pair security 101 source internal destination external
- service-policy type inspect 101
- ex
- interface G0/0
- zone-member security internal
- exit
- interface G0/1
- zone-member security external
- exit
- exit

```
R1(config)#ip access-list extended 101
R1(config-ext-nacl)#permit ip 192.168.0.0 0.0.0.255 any
R1(config-ext-nacl)#ex
R1(config)#class-map type inspect match-all 101
R1(config-cmap)#match access-group name 101
R1(config-cmap)#exit
R1(config)#policy-map type inspect 101
R1(config-pmap)#class type inspect 101
R1(config-pmap-c)#inspect
%No specific protocol configured in class 101 for inspection. All protocols will be inspected
R1(config-pmap-c)#exit
R1(config-pmap)#exit
R1(config)#zone-pair security 101 source internal destination external
R1(config-sec-zone-pair)#service-policy type inspect 101
R1(config-sec-zone-pair)#ex
R1(config)#interface G0/0
R1(config-if)#zone-member security internal
R1(config-if)#exit
R1(config)#interface G0/1
R1(config-if)#zone-member security external
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

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Outputs:

PC B

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=5ms TTL=127
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time=2ms TTL=127

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```

Server

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.0.3

Pinging 192.168.0.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

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